

Abstract Algebra – Dummit & Foote

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Part I

Group Theory

1 Introduction to Groups

1.1 Axioms

If $\star$ is a function on set G such that $\star : G \times G \rightarrow G$, then $\star$ is a **binary operation** on G .

- Associative if $\forall a, b \in G, a \star (b \star c) = (a \star b) \star c$
- Commutative if $\forall a, b \in G, a \star b = b \star a$

A **group** is an ordered pair $(G, \star)$ such that

1. $\star$ is associative
2. Existence of identity: $\exists e \in G$ such that $\forall a \in G, a \star e = e \star a = a$
3. Existence of inverse: $\forall a \in G, \exists a^{-1} \in G$ such that $a \star a^{-1} = a^{-1} \star a = e$

*A group is **Abelian (commutative)** if $\forall a, b \in G, a \star b = b \star a$

E.g. $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ are groups under $+$ with identity $e = 0$ and inverse $a^{-1} = -a$. But $\mathbb{Z} - \{0\}$ is not a group under $\times$ because inverses are not guaranteed to exist.

Let $(G, \star)$ be a group, and $x \in G$. The **order** of x is the smallest positive integer n such that $x^n = 1$, and is denoted $|x|$. If no positive powers of x is 1, $|x| = \infty$.

- In the additive group $\mathbb{Z}/9\mathbb{Z}$, $\bar{6}$ has order 3, since $\bar{6} + \bar{6} + \bar{6} = \bar{18} = \bar{0}$.
- In the multiplicative group $\mathbb{R} - \{0\}$, -1 has order 2 since $(-1)^2 = 1$.

1.2 Dihedral Groups

For each $n \in \mathbb{Z}^+, n \geq 3$, let D_{2n} be the set of symmetries of a regular n -gon. E.g. $s \in D_{2n}$ might be a rotation of $2\pi/n$ around the n -gon.

$|D_{2n}| = 2n$, so we call D_{2n} the **dihedral group of order $2n$** .

Any symmetry can be written in terms of rotations (r) and reflections (s) alone, e.g. for $n = 12$,

$$(sr^9)(sr^6) = s(r^9s)r^6 = s(sr^{-9})r^6 = s^2r^{-3} = r^9$$

S is a set of generators of G if any element of G can be written as a finite product of the elements of S . This is notated:

$$G = \langle S \rangle$$

For the dihedral group, $S = \{s, r\}$. The equations in G that the generators satisfy are called **relations** in G . For D_{2n} , the relations are $r^n = 1$, $s^2 = 1$, and $rs = sr^{-1}$.

If G is generated by S , and there exists a collection of relations $R_1, R_2, \dots, R_m$, then these generators and relations are called the **presentation of G** :

$$G = \langle S \mid R_1, R_2, \dots, R_m \rangle$$

For the dihedral group, $D_{2n} = \langle r, s \mid r^n = s^2 = 1, rs = sr^{-1} \rangle$.

1.3 Symmetric Groups

Let S_Ω be the bijection from Ω to itself (permutation). S_Ω is a group under the function composition $\circ$.

- Call $(S_\Omega, \circ)$ the **symmetric group** on set Ω .
- If $\Omega = \{1, 2, 3, \dots, n\}$, then we denote S_Ω as S_n , the **symmetric group of degree n** .
- $(S_n, \circ)$ has degree $n!$ ($n!$ elements).

We can represent S_n in a cycle decomposition, grouping disjoint cycles. E.g.

$$(a_1 \ a_2 \ \dots \ a_m)(a_{m_1+1} \ a_{m_1+2} \ \dots \ a_{m_2}) \cdots (a_{m_{k-1}+1} \ a_{m_{k-1}+2} \ \dots \ a_{m_k})$$

Meaning that a_1 goes to a_2 , a_{m_1+1} goes to a_{m_1+2} , etc. Values such that $\sigma(a_i) = a_i$ aren't written. For example, for $\sigma \in S_3$ with $\sigma(1) = 1$, $\sigma(2) = 3$, and $\sigma(3) = 2$, the cycle decomposition is

$$(2 \ 3)$$

1.4 Matrix Groups

First, we need to define a field. A field is a set F with 2 binary operators $+$ and $\cdot$ such that

1. $(F, +)$ and $(F - \{0\}, \cdot)$ are Abelian groups
2. The distribution law holds:

$$a \cdot (b + c) = (a \cdot b) + (a \cdot c) \quad \forall a, b, c \in F$$

Define the **general linear group of degree n** :

$$GL_n(F) = \{A \mid A \text{ is a } n \times n \text{ matrix with entries from } F, \text{ and } \det(A) \neq 0\}$$

- Since $\det(A) = 0$, $GL_n(F)$ is closed under matrix multiplication.
- Matrix products are associative.
- $\det(A) \neq 0$ iff. A has an inverse.

Thus, $GL_n(F)$ is a group under matrix multiplication.

1.5 The Quaternion Group

Simple enough:

$$\begin{aligned}
 Q_8 &= \{1, -1, i, -i, j, -j, k, -k\} \\
 1 \cdot a &= a \cdot 1 = a \\
 (-1) \cdot a &= a \cdot (-1) = -a \\
 i \cdot i &= j \cdot j = k \cdot k = -1 \\
 i \cdot j &= k, \quad j \cdot k = i, \quad k \cdot i = j \\
 j \cdot i &= -k, \quad k \cdot j = -i, \quad i \cdot k = -j
 \end{aligned}$$

$(Q_8, \cdot)$ is a non-Abelian group of order 8.

1.6 Homomorphisms and Isomorphisms

Let $(G, \star)$ and $(H, \diamond)$ be groups. A map $\phi : G \rightarrow H$ such that

$$\phi(x \star y) = \phi(x) \diamond \phi(y)$$

is a **homomorphism**.

- “A map that reflects the group structure of the domain and codomain.”

$\phi : G \rightarrow H$ is called an **isomorphism** if

1. ϕ is a homomorphism
2. ϕ is a bijection

Can also say: G and H are isomorphic; G and H are of the same isomorphism type; $G \cong H$.

I.e. G and H are the “same group” except the elements and operators are written differently (think abstract).

If $\phi : G \rightarrow H$ is an isomorphism, then

1. $|G| = |H|$
2. G is Abelian iff. H is Abelian
3. $\forall x \in G, |x| = |\phi(x)|$

1.7 Group Actions

A group action from group G on set A is a map from $G \times A$ to A satisfying

1. $g_1 \cdot (g_2 \cdot a) = (g_1 g_2) \cdot a \quad \forall g_1, g_2 \in G, \forall a \in A$
 2. $1 \cdot a = a \quad \forall a \in A$
-

Let G act on A . For each $g \in G$, define σ_g to be the map

$$\begin{aligned}\sigma_g : A &\rightarrow A \\ \sigma_g(a) &= g \cdot a\end{aligned}$$

1. For each $g \in G$, σ_g is a permutation of A .
2. The map from G to S_A defined by $g \mapsto \sigma_g$ is a homomorphism.

I.e. every $g \in G$ acts as a permutation of A in a way consistent with the group operation in G .

Group actions treat $g \in G$ to be functions, sort of.

2 Subgroups

A subset H of a group H is a **subgroup** of G if

1. $H \neq \emptyset$
2. H is closed under products and inverses. I.e. $x, y \in H$ implies $x^{-1} \in H$ and $xy \in H$.

This is denoted $H \leq G$, or if $H \neq G$, $H < G$.

I.e. a subgroup is a subset of G that is also a group w.r.t the binary operation defined on G . A subgroup must thus contain the identity operator.

Every G has two subgroups: $H = G$ and $H = \{1\}$ (trivial subgroup).

A simpler subgroup criterion: a subset H of a group G is a subgroup of G iff.

1. $H \neq \emptyset$
2. $\forall x, y \in H, xy^{-1} \in H$

2.1 Centralizers and Normalizers, Stabilizers and Kernels

Let A be a subset of G .

The **centralizer of A in G** is the set of elements of G that commute with every element of A .

$$C_G(A) = \{g \in G \mid gag^{-1} = a, \forall a \in A\}$$

- For an Abelian group G , $C_G(A) = G, \forall A \subseteq G$.
- For the quaternion group, $C_{Q_8}(\{i\}) = \{\pm 1, \pm i\}$

The **center of G** is the set of elements of G commuting with every element of G .

$$Z(G) = C_G(G) = \{g \in G \mid gx = xg, \forall x \in G\}$$

The **normalizer of A in G** is the set of elements of G that commutes with the entire set A .

$$N_G(A) = \{g \in G \mid gAg^{-1} = A\}$$

This is a looser criterion than the centralizer, since order within A doesn't matter. Formally, if $g \in C_G(A)$, then $gag^{-1} = a$ for all $a \in A$, so $C_G(A) \leq N_G(A)$.

For an Abelian group G , which has the commutation property already, $C_G(A) = N_G(A) = Z(G) = G$, for any subset $A \subseteq G$.

If G is a group action on set S , and $s \in S$, the **stabilizer of s in G** is

$$G_s = \{g \in G \mid g \cdot s = s\}$$

- $1 \in G_s$
- If $y \in G_s$, then $y^{-1} \in G_s$.

$$\begin{aligned} s &= 1 \cdot s = (y^{-1}y) \cdot s \\ &= y^{-1} \cdot (y \cdot s) \\ &= y^{-1} \cdot s \end{aligned}$$

- If $x, y \in G_s$, then $xy \in G_s$.

$$\begin{aligned} (xy) \cdot s &= x \cdot (y \cdot s) \\ &= x \cdot s \\ &= s \end{aligned}$$

Thus G_s is a subgroup of the group action G , or $G_s \leq G$.

The **kernel of the action of G on set S** is

$$\{g \in G \mid g \cdot s = s, \forall s \in S\}$$

2.2 Cyclic Subgroups

A possible way to generate a subgroup H of G from an element $x \in G$ is to form the set $H = \{x^n \mid n \in \mathbb{Z}\}$. Guarantees closure under products and inverses.

A subgroup $H \subseteq G$ is **cyclic** if $\exists x \in H$ such that $H = \{x^n \mid n \in \mathbb{Z}\}$. I.e. H can be generated by 1 element. Notated $H = \langle x \rangle$. Cyclic groups are Abelian, by properties of the exponent.

- E.g. If $H = (\mathbb{Z}, +)$, then $H = \langle 1 \rangle$.
- E.g. If H is the subgroup of all rotations of an n -gon, then $H = \langle r \rangle$.

2.2.1 Theorems

1. If $H = \langle x \rangle$, then $|H| = |x|$.
2. Let $x \in G$. For $m, n \in \mathbb{Z}$, if $x^n = x^m = 1$, then $x^d = 1$, where $d = \gcd(m, n)$.
3. Any 2 cyclic groups of the same order are isomorphic.

- If $n \in \mathbb{Z}^+$ and $\langle x \rangle, \langle y \rangle$ are cyclic groups of order n , the following map is an isomorphism.

$$\begin{aligned}\phi : \langle x \rangle &\rightarrow \langle y \rangle \\ x^k &\mapsto y^k\end{aligned}$$

- If $\langle x \rangle$ is an infinite cyclic group, the following map is a homomorphism. Thus for each $n \in \mathbb{Z}^+$, let Z_n be the cyclic group of order n .

$$\begin{aligned}\phi : \mathbb{Z} &\rightarrow \langle x \rangle \\ k &\mapsto x^k\end{aligned}$$

4. Let G be a map, $x \in G$, and $a \in \mathbb{Z} - \{0\}$.

- If $|x| = \infty$, then $|x^a| = \infty$
- If $|x| = n < \infty$, then $|x^a| = n / \gcd(n, a)$

5. Let $H = \langle x \rangle$.

- If $|x| = \infty$, then $H = \langle x \rangle$ iff. $a = \pm 1$.
- If $|x| = n < \infty$, then $H = \langle x^a \rangle$ iff. $\gcd(a, n) = 1$. The number of generators of H is $\phi(n)$.
- E.g. $\bar{a}$ generates $\mathbb{Z}/n\mathbb{Z}$ iff. $(\bar{a}, n) = 1$. So $\bar{1}, \bar{5}, \bar{7}, \bar{11}$ are generators of $\mathbb{Z}/12\mathbb{Z}$, and these are all the $\phi(12) = 4$ generators.

6. Let $H = \langle x \rangle$ be a cyclic group. Then

- Every subgroup of H is cyclic.
- If $|H| = \infty$, then for any non-negative and distinct $a, b \in \mathbb{Z}$, $\langle x^a \rangle = \langle x^b \rangle$.
- If $|H| = n < \infty$, then for each $a \in \mathbb{Z}^+$ such that $a|n$, there's a unique subgroup of order a . This is the cyclic group $\langle x^d \rangle$, where $d = n/a$.

2.3 Subgroups Generated by Subsets

If $\mathcal{A}$ is any nonempty collection of subgroups of G , the intersection of all $\mathcal{A}$ is also a subgroup of G .

$$K = \bigcap_{H \in \mathcal{A}} H \quad \text{is a group if} \quad H \leq G$$

If A is a subset of the group G , let the **subgroup of G generated by A** be

$$\langle A \rangle = \bigcap_{A \subseteq H, H \leq G} H$$

I.e. the intersection of all subgroups of G containing A . Equivalently,

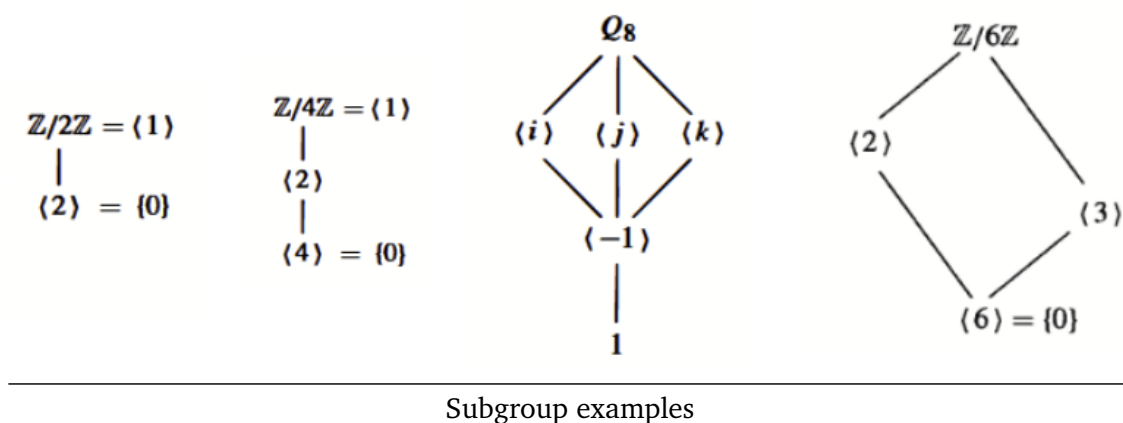
$$\langle A \rangle = \{a_1^{\alpha_1} a_2^{\alpha_2} \cdots a_n^{\alpha_n} \mid \forall i, a_i \in A, \alpha_i \in \mathbb{Z}, a_i \neq a_{i+1} \text{ and } n \in \mathbb{Z}^+\}$$

An Abelian group G simplifies this a bit:

$$\langle A \rangle = \{a_1^{\alpha_1} a_2^{\alpha_2} \cdots a_n^{\alpha_n} \mid \forall i \text{ and } \alpha_i \in \mathbb{Z}\}$$

- Without an Abelian structure, determining the generated subgroup becomes difficult. E.g. if $G = D_8$, let r, s be the generators. Let $a = s, b = rs$ so that $G = \langle a, b \rangle$. Thus G is generated by 2 elements of order 2, but has order 8.
- Or for $G = S_n = \langle (1\ 2), (1\ 2\ 3\ \dots\ n) \rangle$. Generated by elements of order 2 and n , but has order $n!$

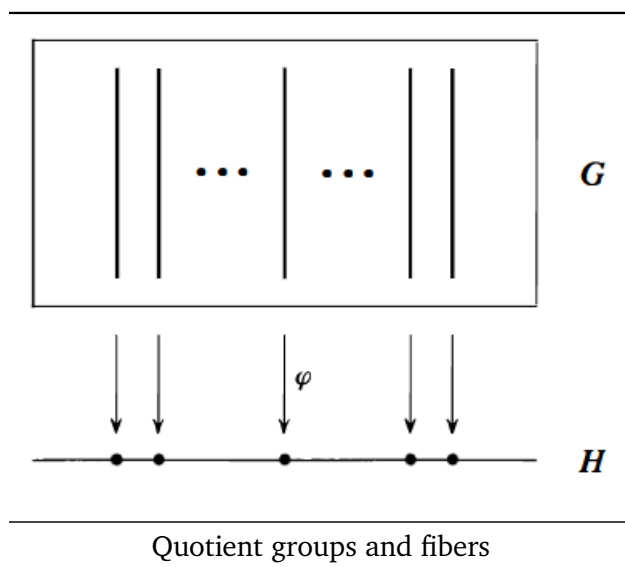
Lattice diagrams can help illuminate the structure of groups and subgroups. Build up from $\{0\}$ at the bottom, and draw subgroups of increasingly higher order upwards. This is helpful to see, for example, the smallest subgroup of G , $\langle H, K \rangle$, containing two smaller subgroups $H \leq G$ and $K \leq G$.



3 Quotient Groups and Homomorphisms

3.1 Definitions

Quotient groups are a way to obtain a smaller group from a group G .



$\phi : G \rightarrow H$ is a homomorphism from G to H , i.e. $\phi(xy) = \phi(x)\phi(y)$ for $x, y \in G$. Vertical lines above each point $h \in H$ represent the **fiber of ϕ over h** .

Denote the **kernel of ϕ** as the set

$$\ker \phi = \{g \in G \mid \phi(g) = 1_H\}$$

Let K be the kernel. The **quotient group/factor group** G/K (" $G \bmod K$ ") is a group whose elements are the fibers of ϕ .

For any subgroup $N \leq G$ and $g \in G$, define

$$\text{Left coset of } N \text{ in } G: gN = \{gn \mid n \in N\}$$

$$\text{Right coset of } N \text{ in } G: Ng = \{ng \mid n \in N\}$$

Call any element of a coset the **representative** of that coset. The fibers of a homomorphism $\phi : G \rightarrow H$ are the left or right cosets of the kernel (it may help to think of this graphically). The elements of G/K are the left cosets gK or the right cosets Kg with $g \in G$.

3.2 Theorems

3.2.1 Quotient Groups from Kernel

Theorem 1: Let $\phi : G \rightarrow H$ be a homomorphism. Then

1. $\phi(1_G) = 1_H$
2. $\phi(g^{-1}) = \phi(g)^{-1}$
3. $\phi(g^n) = \phi(g)^n$
4. Kernel of ϕ is a subgroup of G : $\ker \phi \leq G$
5. Image of G under ϕ is a subgroup of H : $\text{im}(\phi) \leq H$

Proof: For (1), $\phi(1_G) = \phi(1_G 1_G) = 1_H 1_H$. For (2), $\phi(g)\phi(g^{-1}) = \phi(gg^{-1}) = \phi(1_G) = 1_H$. Then (3) follows from induction.

For (4), note that $1_G \in \ker \phi$, so $\ker \phi$ is nonempty. Let $x, y \in \ker \phi$. Then $\ker \phi$ satisfies the subgroup criterion:

$$\phi(xy^{-1}) = \phi(x)\phi(y)^{-1} = 1_H$$

Similarly for (5), note that $\phi(1_G) = 1_H$, so the identity of H lies in $\text{im}(\phi)$. If $x, y \in \text{im}(\phi)$ with $\phi(a) = x$ and $\phi(b) = y$, then $\text{im}(\phi)$ satisfies the subgroup criterion:

$$\begin{aligned} xy^{-1} &= \phi(a)\phi(b)^{-1} = \phi(ab^{-1}) \\ xy^{-1} &\in \text{im}(\phi) \end{aligned}$$

Definition: Let $\phi : G \rightarrow H$ with kernel K . Then the **quotient group** G/K is the group whose elements are fibers of ϕ with group operations defined such that: if X is the fiber above a and Y is the fiber above b , then $X \cdot Y$ is the fiber above ab .

Thus $K \in G/K$. Other elements of G/K are “translations” of K .

Theorem 2: Let $\phi : G \rightarrow H$ be a homomorphism with kernel K . Let $X \in G/K$ be a fiber above a , i.e. $X = \phi^{-1}(a)$. Then

- 1) $\forall u \in X, X = \{uk \mid k \in K\}$
- 2) $\forall u \in X, X = \{ku \mid k \in K\}$

Proof: First show $uK \subseteq X$, then $X \subseteq uK$. Similarly for Ku .

What this shows is that the fibers of a homomorphism are the left (or right) cosets of the kernel. Furthermore, the specific value of u doesn't matter, as long as $u \in X$. This is why u is called the “**representative**.”

Theorem 3: Let K be the kernel of $\phi : G \rightarrow H$. The set of left cosets of G with operation defined as

$$uK \cdot vK = (uv)K$$

form the quotient group G/K . Similarly for right cosets.

E.g. for $\phi : \mathbb{Z} \rightarrow \mathbb{Z}_n$, the cosets are $a + n\mathbb{Z}$ and the kernel is $n\mathbb{Z}$. Thus the fibers $\mathbb{Z}/n\mathbb{Z}$ form a group.

In essence, to find a quotient group:

1. Find homomorphism ϕ from group G .
2. Find kernel K of homomorphism.
3. G/K , the fibers of ϕ , is a group

3.2.2 Quotient Groups from Normal Subgroups

We can form a group from a subgroup N iff. N is the kernel of some homomorphism. To determine whether N is a kernel of such a homomorphism, check whether N is a normal subgroup: $N \trianglelefteq G$.

Theorem: For a subgroup N of G , i.e. $N \leq G$, the following are equivalent conditions.

1. $N \trianglelefteq G$
2. $N_G(N) = G$
3. $gN = Ng, \quad \forall g \in G$
4. $gNg^{-1} \subseteq N, \quad \forall g \in G$

Theorem: $N \leq G$ is normal iff. it's the kernel of some homomorphism.

Proof: If N is the kernel of some homomorphism, its left and right cosets are equal (both are fibers of ϕ). By the previous theorem, if $gN = Ng$ for all $g \in G$ is equivalent to $N \trianglelefteq G$.

In the opposite direction, assume $N \trianglelefteq G$ and let $\pi : G \rightarrow G/N$ with $\pi(g) = gN$. Then

$$\pi(g_1g_2) = g_1g_2N = (g_1N)(g_2N) = \pi(g_1)\pi(g_2)$$

so π is a homomorphism. The kernel is $\ker \pi = \{g \in G \mid gN = 1N\} = N$, the normal subgroup.

For $N \trianglelefteq G$, the homomorphism $\pi : G \rightarrow G/N$ defined by $\pi(g) = gN$ is the **natural projection homomorphism**.

Thus an easy way to see whether a subgroup $N \leq G$ is a kernel of some homomorphism is to check whether $N_g(N) = G$. Then we can find the quotient group G/N .

Note that in an Abelian group, any subgroup $N \leq G$ is normal, so any $N \leq G$ can be used to create a quotient group G/N .

3.3 Lagrange's Theorem

3.3.1 Statement

Theorem: The left cosets of $N \leq G$ form a partition of G , and $uN = vN$ iff. u, v are representatives of the same coset.

Lagrange's Theorem: If G is a finite group and $H \leq G$, then $|H|$ divides $|G|$. The number of left cosets of H in G is $|G|/|H|$.

Proof: Let $|H| = n$ and k be the number of left cosets of H in G . For some $g \in G$, define $\phi : H \rightarrow gH$ with $\phi(h) = gh$. ϕ is both injective and surjective, so $|gH| = |H| = n$. From the previous theorem, the cosets partition G , so $|G| = k|H| = kn$. Thus $|G|/|H| = k$.

If $H \leq G$, the number of left cosets of H in G is called the **index** of H in G , denoted $|G : H|$.

- E.g. $\{0\} \in \mathbb{Z}$ has infinite index.
- E.g. $\langle n \rangle \in \mathbb{Z}$ has index n .

3.3.2 Corollaries

Corollary 1: If $x \in G$ and $|G|$ is finite, then $|x|$ divides $|G|$, i.e. $x^{|G|} = 1$.

Proof: $|x| = |\langle x \rangle|$. Let $H = \langle x \rangle$. Then $H \leq G$ so $|H| = |x|$ divides $|G|$.

Corollary 2: If $|G|$ is prime, then G is cyclic and $G \cong Z_p$.

Proof: Let $x \in G$ and $x \neq 1$. Then $|\langle x \rangle| \geq 1$. $\langle x \rangle \in G$, so $|\langle x \rangle|$ divides $|G|$. But $|G|$ is prime, so $|G| = |\langle x \rangle|$. Hence G is cyclic with order p .

Corollary 3: Let $H \leq G$ and $|G : H| = 2$. Then $H \trianglelefteq G$.

Proof: Let $g \in G - H$. The two left cosets of H in G are $1H$ and gH . They partition G , so $gH = G - H$. Similarly, $Hg = g - H$. Then $gH = Hg$, so H must be a normal subgroup of G .

- E.g. $\langle i \rangle = \{i, -1, -i, +1\}$, which has order 4. Q_8 has order 8, so $|Q_8 : \langle i \rangle| = 2$. Thus $\langle i \rangle \trianglelefteq Q_8$.

3.3.3 Cauchy's Theorem and Sylow's Theorem

The converse to Lagrange's theorem isn't true generally: just because n divides G doesn't mean there exists $H \leq G$ with $|H| = n$.

There are some partial converses:

Cauchy's Theorem: If G is finite, and there exists a prime p dividing $|G|$, then G has an element of order p .

Sylow's Theorem: If G is a finite group of order $p^\alpha m$, where p is prime and doesn't divide m , then G has a subgroup of order p^α .

3.3.4 Applications: Fermat's Little Theorem, Euler's Theorem

Lagrange's theorem provides an elegant way to prove these (seemingly number-theoretic) theorems.

Fermat's Little Theorem: If p is prime, then $a^p \equiv a \pmod{p}$ for all $a \in \mathbb{Z}$.

Proof: Define the multiplicative group modulo p , i.e. $(\mathbb{Z}/p\mathbb{Z})^\times$. For all $a \in \mathbb{Z}$ (with operation being multiplication mod p), $\langle a \rangle \in (\mathbb{Z}/p\mathbb{Z})^\times$. Thus $|a| = |\langle a \rangle|$ must divide $p - 1$, so $a^{p-1} \equiv 1 \pmod{p}$.

Euler's Theorem: For all a relatively prime to n , $a^{\phi(n)} \equiv 1 \pmod{n}$, where ϕ is the Euler totient function.

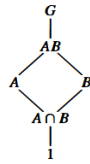
Proof: Define the multiplicative group mod n , $(\mathbb{Z}/n\mathbb{Z})^\times$. Also define the multiplicative group mod n of elements *relatively prime* to n , $(\mathbb{Z}/n\mathbb{Z})^*$. Then $|(\mathbb{Z}/n\mathbb{Z})^*| = \phi(n)$. Let a be relatively prime to n , so $a \in (\mathbb{Z}/n\mathbb{Z})^*$. Then all powers of a are also relatively prime to n , so $\langle a \rangle \in (\mathbb{Z}/n\mathbb{Z})^*$.

Use Lagrange's theorem to see that $|a| = |\langle a \rangle|$ must divide $\phi(n)$. Thus $a^{\phi(n)} \equiv 1 \pmod{n}$.

3.4 Isomorphism Theorems

1st Isomorphism Theorem: If $\phi : G \rightarrow H$ is a homomorphism, then $\ker \phi \trianglelefteq G$ and $G/\ker \phi \cong \phi(G)$.

2nd Isomorphism Theorem: Let $A, B \leq G$ and $A \leq N_G(B)$. Then $AB \leq G$, $B \trianglelefteq AB$, $A \cap B \trianglelefteq A$, and $AB/B \cong A/A \cap B$.



Lattice diagram of 2nd isomorphism theorem

3rd Isomorphism Theorem: Let $H, K \trianglelefteq G$ with $H \leq K$. Then $K/H \trianglelefteq G/H$ and

$$(G/H)/(K/H) \cong G/K$$

4th Isomorphism Theorem: Let $N \trianglelefteq G$. There's a bijection from the set of subgroups $A \leq G$ containing N onto the set of subgroups A/N of G/N .

I.e. every subgroup of G/N is of the form A/N for some $A \leq G$ containing N .

3.5 The Hölder Program

3.5.1 Composition Series

Call G a **simple group** if $|G| > 1$ and the only normal subgroups of G are 1 and G .

The sequence

$$1 = N_0 \leq N_1 \leq N_2 \leq \cdots \leq N_{k-1} \leq N_k = G$$

is called a **composition series** if $N_i \trianglelefteq N_{i+1}$ and N_{i+1}/N_i is a simple group. The quotient groups N_{i+1}/N_i are **composition factors** of G .

E.g. for the dihedral group D_8 ,

$$\begin{aligned} 1 \trianglelefteq \langle s \rangle \trianglelefteq \langle s, r^2 \rangle \trianglelefteq D_8 \\ 1 \trianglelefteq \langle r^2 \rangle \trianglelefteq \langle r \rangle \trianglelefteq D_8 \end{aligned}$$

3.5.2 Hölder Program

Jordan-Hölder Theorem: Let G be a finite group with $|G| > 1$. Then G has a composition series, and the composition factors are unique up to a permutation.

This suggests the **Hölder program**:

1. Classify all finite groups
2. Find all ways of combining two groups into one.

Theorem: There are 18 infinite simple group families and 26 simple groups not in those families (sporadic simple groups), such that every finite simple group is isomorphic to 1 in this list.

Feit-Thompson Theorem: If G is a simple group of odd order, then $G \cong Z_p$ for some prime p .

Proof: Both are trivial.

We call G **solvable** if it has the composition

$$1 = G_0 \trianglelefteq G_1 \trianglelefteq G_2 \trianglelefteq \cdots \trianglelefteq G_s = G$$

where G_{i+1}/G_i is Abelian.

3.6 The Alternating Group A_n

Call a 2-cycle $(i\ j)$ a **transposition**. Every element of S_n can be written as a product of transpositions.

Define the polynomial Δ and the permutation σ such that

$$\begin{aligned} \Delta &= \prod_{1 \leq i < j \leq n} (x_i - x_j) \\ \sigma(\Delta) &= \prod_{1 \leq i < j \leq n} (x_{\sigma(i)} - x_{\sigma(j)}) \end{aligned}$$

Since σ is a bijection, only a few terms $(x_{\sigma(i)} - x_{\sigma(j)})$ get their signs switched, so $\sigma(\Delta) = \pm\Delta$. Define ϵ such that

$$\epsilon(\sigma) = \begin{cases} +1 & \sigma(\Delta) = \Delta \\ -1 & \sigma(\Delta) = -\Delta \end{cases}$$

- E.g. for $\sigma = (1\ 2\ 3\ 4)$, we have $(1\ 2\ 3\ 4)(\Delta) = (-1)^3(\Delta)$, so $\epsilon((1\ 2\ 3\ 4)) = -1$. This is **not** the Levi-Civita symbol.
- $\epsilon(\sigma)$ is called the **sign** of σ .
- σ is an even permutation if $\epsilon(\sigma) = +1$, odd if $\epsilon(\sigma) = -1$.
- The map $\epsilon : S_n \rightarrow \{\pm 1\}$ is a homomorphism.

Theorem: transpositions $(i\ j)$ are all odd permutations.

Proof: Note that $\epsilon((1\ 2)) = -1$ since it only flips $(x_1 - x_2)$. For a general $(i\ j)$, let λ be the permutation exchanging 1 and i , 2 and j . Then $(i\ j) = \lambda(1\ 2)\lambda$. ϵ is a homomorphism, so

$$\epsilon((i\ j)) = \epsilon(\lambda(1\ 2)\lambda) = \epsilon(\lambda)^2\epsilon((1\ 2)) = -1$$

Define A_n , the **alternating group of degree n** , to be the kernel of ϵ . I.e. the set of even permutations. $S_n - A_n$ is the set of odd permutations.

By the 1st isomorphism theorem, $S_n/\ker \epsilon \cong \{\pm 1\}$, and by Lagrange's theorem, $|A_n| = n!/2$. A_n will become useful in the study of polynomials.

4 Group Actions

4.1 Definitions

Let G be a group acting on set A . For all $g \in G$, the map $\sigma_g : A \rightarrow A$ defined by $\sigma_g : a \mapsto g \cdot a$ is a permutation of A .

For each action of G on A , there's an associated homomorphism $\phi : G \rightarrow S_A$ defined by $\phi(g) = \sigma_g$, called the **permutation representation** associated with a given action.

- Kernel of action: $\{g \in G \mid g \cdot a = a, \forall a \in A\}$
 - Stabilizer of a in G : $G_a = \{g \in G \mid g \cdot a = a\}$
 - An action is **faithful** if the kernel is the identity.
-

A **permutation representation** of G is any homomorphism of G into the symmetric group S_A for some nonempty set A .

- Given any nonempty set A and homomorphism $\phi : G \rightarrow S_A$, the action of G on A can be $g \cdot a = \phi(g)(a)$. The kernel of this action is the same as the kernel of ϕ .

Let G act on set A . The relation on A defined by $a \sim b$ iff. $a = g \cdot b$ for some $g \in G$ is an **equivalence relation**. The **equivalence class** is $\mathcal{C}_a = \{g \cdot a \mid g \in G\}$. Alternatively, the **orbit of G containing a** .

- The orbit of a : set of all elements a can move to under a group action.
- $|\mathcal{C}_a| = |G : G_a|$, index of stabilizer a
- Action of G on A is **transitive** if there's only 1 orbit. I.e. for all $a, b \in A$, there exists $g \in G$ such that $a = g \cdot b$.

Cayley's Theorem: Every group is isomorphic to a subgroup of some symmetric group. If G has order n , then G is isomorphic to a subgroup of S_n .

Proof: Let G be a group, and let A be the set of left cosets of $1 \in G$. Let G act by left multiplication on A . The homomorphism $\pi : G \rightarrow S_G$ is that of a group action. The kernel of π is

$$\begin{aligned} \ker \pi &= \{g \in G \mid gx1 = x1 \quad \forall x \in G\} \\ &= \{1\} \end{aligned}$$

Thus π is an isomorphism.

4.2 Groups Acting on Themselves by Conjugation

4.2.1 Definitions

Let G act on $A = G$ through conjugation, such that

$$g \cdot a = gag^{-1}, \quad \forall g, a \in G$$

$a, b \in G$ are conjugate in G if there exists a $g \in G$ such that $b = g \cdot a = gag^{-1}$. I.e. if a and b are in the same orbit of G acting on itself by conjugation. These orbits are called the **conjugacy classes** of G .

- For an Abelian group, the conjugacy class of each $g \in G$ is $\{g\}$.
- Subsets $S, T \subseteq G$ are conjugate in G if there exists a $g \in G$ such that $T = gSg^{-1}$. I.e. if S, T are in the same orbit of G acting on itself by conjugation.

For $S \subseteq G$, if G acts on S with $g \cdot a = gag^{-1}, \forall a \in S$, the number of elements in the orbit/equivalence class containing S is $|G : G_S|$.

$$G_S = \{g \in G \mid gSg^{-1} = S\} = N_G(S)$$

- The number of conjugates of a subset $S \subseteq G$ is the index of the normalizer of S , i.e. $|G : N_G(S)|$.
- The number of conjugates of an element $s \in G$ is the index of the centralizer of s , i.e. $|G : C_G(s)|$.

The Class Equation: Let G be a finite group, and let $g_1, g_2, \dots, g_r$ be the representatives of distinct conjugacy classes of G not contained in the center $Z(G)$. Then

$$|G| = |Z(G)| + \sum_{i=1}^r |G : C_G(g_i)|$$

Corollary: If P is a group of prime power p^α for some $\alpha \geq 1$, then P has a nontrivial center $Z(P) \neq 1$.

Proof: By the class equation,

$$|P| = |Z(P)| + \sum_{i=1}^r |P : C_P(g_i)|$$

and by assumption, $C_P(g_i)$ is non-central, so $|C_P(g_i)| \neq |P|$. p divides $|P : C_P(g_i)|$, so p must divide $|Z(P)|$ and $|Z(P)| \neq 1$.

4.2.2 Conjugacy in Symmetric Groups

Let $\sigma, \tau \in S_n$ and let σ have the cycle decomposition $(a_1 \ a_2 \ \dots \ a_{k_1})(b_1 \ b_2 \ \dots \ b_{k_2}) \dots$. Then $\tau\sigma\tau^{-1}$ has the cycle decomposition

$$(\tau(a_1) \ \tau(a_2) \ \dots \ \tau(a_{k_1}))(\tau(b_1) \ \tau(b_2) \ \dots \ \tau(b_{k_2})) \dots$$

Proof: If $\sigma(i) = j$, then $\tau\sigma\tau^{-1}(\tau(i)) = \tau(j)$. Thus if i, j appears in the cycle decomposition of σ , then $\tau(i), \tau(j)$ appears in the cycle decomposition of τ .

If $\sigma \in S_n$ is the product of disjoint cycles of length $n_1, n_2, \dots, n_r$ with $n_1 \leq n_2 \leq \dots \leq n_r$, then $n_1, n_2, \dots, n_r$ are the **cycle type** of σ . Two elements of S_n are conjugate iff. they have the same cycle type.

The number of partitions of n equals the number of conjugacy classes of S_n . E.g. for $n = 5$,

Partition of 5	Representative of Conjugacy Class
1, 1, 1, 1, 1	1
1, 1, 1, 2	(1 2)
1, 1, 3	(1 2 3)
1, 4	(1 2 3 4)
5	(1 2 3 4 5)
1, 2, 2	(1 2)(3 4)
2, 3	(1 2)(3 4 5)

4.3 Automorphisms

An isomorphism of G onto itself is called an **automorphism**. The set of all automorphisms of G is $\text{Aut}(G)$.

Let $H \leq G$. Then G acts by conjugation on H as an automorphism of H . I.e. conjugation by $g \in G$ is an automorphism of H . The permutation representation of the homomorphism is $\phi : G \rightarrow \text{Aut}(H)$ with kernel $C_G(H)$.

$$h \mapsto ghg^{-1}, \quad \forall h \in H$$

Let $g \in G$. Conjugation by g is an **inner automorphism** of G , and the subgroup of $\text{Aut}(G)$ containing all inner automorphisms of G is called $\text{Inn}(G)$.

- Note that $\text{Inn}(G) \leq \text{Aut}(G)$, so $\text{Inn}(G) \cong G/Z(G)$.

A subgroup $H \leq G$ is **characteristic in G** , denoted $H \text{ char } G$, if every automorphism of G maps H to itself: $\sigma(H) = H$ for all $\sigma \in \text{Aut}(G)$.

- Characteristic subgroups are normal.
- Although normality isn't transitive, if $K \text{ char } H$ and $H \trianglelefteq G$, then $K \trianglelefteq G$.
- Characteristic subgroups are like “strongly normal” subgroups.

Theorem: For any $H \leq G$, $N_G(H)/C_G(H)$ is isomorphic to a subgroup of $\text{Aut}(H)$. If $H = G$, then this becomes that $G/Z(G)$ is isomorphic to a subgroup of $\text{Aut}(G)$.

Proof: Note that $H \trianglelefteq N_G(H)$. Conjugation by $g \in G$ is an automorphism of H with permutation representation $\phi : N_G(H) \rightarrow \text{Aut}(H)$ with kernel $C_G(H)$. By the 1st isomorphism theorem,

$$N_G(H)/C_G(H) \cong \text{Subgroup of } \text{Aut}(H)$$

Theorem: The automorphism group of cycle groups of order n , Z_n , is isomorphic to $(\mathbb{Z}/n\mathbb{Z})^\times$, an Abelian group of order $\phi(n)$.

Proof: Let $Z_n = \langle x \rangle$. If $\psi \in \text{Aut}(Z_n)$, then $\psi(x) = x^a$ for some $a \in \mathbb{Z}$. Further, $|x| = n$ so a is defined mod n . We seek to show a bijection between Z_n and $(\mathbb{Z}/n\mathbb{Z})^\times$. Define the homomorphism (it can be shown that $\Psi(\psi_a \circ \psi_b) = \Psi(\psi_a) \circ \Psi(\psi_b)$)

$$\begin{aligned} \Psi : \text{Aut}(Z_n) &\rightarrow (\mathbb{Z}/n\mathbb{Z})^\times \\ \psi_a &\mapsto a \pmod{n}, \quad \psi_a \in \Psi \end{aligned}$$

This map is surjective. It is also injective, thus Ψ is an isomorphism.

4.4 Sylow Theorems

4.4.1 Statement

Let p be a prime. A group of order p^α for some $\alpha \geq 1$ is called a p -group. A subgroup of G that is a p -group is called a p -subgroup.

If $|G| = p^\alpha m$ where $p \nmid m$. A subgroup of order p^α (i.e. max subgroup of size that's a power of p) is called a **Sylow p -subgroup of G** .

- Set of Sylow p -subgroups of G : $\text{Syl}_p(G)$
- Number of Sylow p -subgroups of G : $n_p(G)$

Sylow Theorems: (partial converse to Lagrange's Theorem) Let $|G| = p^\alpha m$, where p is prime and $p \nmid m$. Then

1. $\text{Syl}_p(G) \neq \emptyset$
 2. If P is a Sylow p -subgroup of G , and Q is any p -subgroup of G , then there exists $g \in G$ such that $Q \leq gPg^{-1}$. Further, any 2 Sylow p -subgroups of G are conjugate in G .
 3. The number of Sylow p -subgroups of G is of the form $1 + kp$, i.e. $n_p \equiv 1 \pmod{p}$. Further, $n_p = |G : N_G(P)|$ for a Sylow p -subgroup P . Thus n_p divides m .
-

If P is a Sylow p -subgroup of G , the following are equivalent:

- 1) P is the unique Sylow p -subgroup of G , $n_p = 1$
- 2) $P \trianglelefteq G$
- 3) P char G
- 4) All subgroups generated by elements of power p^n for some n are p -groups.

4.4.2 Applications

Theorem: If $|G| = pq$ for prime p, q with $p < q$, let $P \in \text{Syl}_p(G)$ and $Q \in \text{Syl}_q(G)$. Then $Q \trianglelefteq G$. If $P \trianglelefteq G$ as well, then G is cyclic.

Proof: (1) Since $Q \in \text{Syl}_q(G)$, $n_q = 1 + kq$ for some $k \geq 0$. And n_q divides p , so $p < q$ and $p = \text{prime}$ enforces $k = 0$. Thus $n_q = 1$, and this is equivalent to $Q \trianglelefteq G$.

Now since $P \in \text{Syl}_p(G)$, $n_p = 1 + kp$ for some $k \geq 0$. Because n_p divides q , either $n_p = 1$ or $n_p = q$. If $n_p = q$, then $q \equiv 1 \pmod{p}$. If $n_p = 1$, then q is not $1 \pmod{p}$, and $P \trianglelefteq G$.

- (2) If $P \trianglelefteq G$, then $n_p = 1$ so q is not $1 \pmod{p}$. Since P, Q have prime power, they're cyclic: $P = \langle x \rangle$ and $Q = \langle y \rangle$. If $P \trianglelefteq G$, $N_G(P) = G$, so $G/C_G(P) \cong \text{Subgroup of } \text{Aut}(P) = \text{Subgroup of } Z_p$. This has order $(p - 1)$, so by Lagrange's theorem,

$$\frac{|G|}{|C_G(P)|} = |Z_p| = p - 1$$

Neither p, q divide $p - 1$, so $G = C_G(P)$. Thus for $x \in P \leq Z_G(P)$, x commutes with all y , and $|xy| = pq$ or $G \cong Z_{pq}$.

E.g. Groups of Order 30: We show that all groups of order 30 have a normal subgroup isomorphic to Z_{15} .

Sketch: Any subgroup of order 15 is normal, since it has index 2. Also cyclic by previous result, since $15 = 3 \cdot 5$ and $3 \nmid (5 - 1)$. So we just show that there exists a subgroup of order 15.

Let $P \in \text{Syl}_p(G)$ and $Q \in \text{Syl}_q(G)$. Assume neither are normal. The possibilities then are $n_5 = 6$, $n_3 = 10$, not $n_5 = n_3 = 1$. There are $4 \cdot 6 = 24$ non-identity elements of order 5, and $2 \cdot 10$ non-identity elements of order 3. Total 44, yet $|G| = 30$, impossible. Thus at least 1 of P, Q must be normal, and

$$|PQ| = \frac{|P||Q|}{|P \cap Q|} = 15$$

5 Direct and Semidirect Products and Abelian Groups

5.1 Direct Products

The **direct product** $G_1 \times G_2 \times \cdots \times G_n$ of groups G_i , $1 \leq i \leq n$, with operations $\star_1, \star_2, \dots, \star_n$, is the set of n -tuples $(g_1, g_2, \dots, g_n)$ where $g_i \in G_i$ and the operation is defined componentwise:

$$(g_1, g_2, \dots, g_n) \star (h_1, h_2, \dots, h_n) = (g_1 \star_1 h_1, g_2 \star_2 h_2, \dots, g_n \star_n h_n)$$

And likewise for infinitely many n . For example, if $G_i = (\mathbb{R}, +)$, then $\mathbb{R} \times \mathbb{R} \times \cdots \times \mathbb{R}$ is the Euclidean n -space $\mathbb{R}^n$ such that

$$(a_1, a_2, \dots, a_n)(b_1, b_2, \dots, b_n) = (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n)$$

- If $G_1, G_2, \dots, G_n$ are groups, their direct product is a group of order $|G_1||G_2| \cdots |G_n|$.
- The order of an element $x_1 x_2 \dots x_n \in G$ is $\text{lcm}(|x_1|, |x_2|, \dots, |x_n|)$.
- For a fixed i , the set of elements of G with 1_{G_j} in the j -th position for all $i \neq j$ is a subgroup of G isomorphic to G_i .

$$G_i \cong \{(1, 1, \dots, 1, g_i, 1, \dots, 1) \mid g_i \in G_i\}$$

5.2 Fundamental Theorem of Finitely-Generated Abelian Groups

G is **finitely generated** if there's a finite subset A of G such that $G = \langle A \rangle$. For each $r \in \mathbb{Z}$ with $r \geq 0$, let $\mathbb{Z}^r = \mathbb{Z} \times \cdots \times \mathbb{Z}$ be the direct product of r $\mathbb{Z}$'s. Then $\mathbb{Z}^r$ is the **free Abelian group of rank r** .

Fundamental Theorem of Finitely Generated Abelian Groups: Let G be a finitely generated Abelian group. Then

$$G \cong \mathbb{Z}^r \times Z_{n_1} \times Z_{n_2} \times \cdots \times Z_{n_s}$$

for integers $r, n_1, n_2, \dots, n_s$ satisfying

1. $r \geq 0$ and $n_i \geq 2$ for all i .
2. n_{i+1} divides n_i for all applicable i .

The decomposition of G in this way is unique, called the **invariant factor decomposition** of G . r is the **free rank/Betti number** of G . Integers $n_1, n_2, \dots, n_s$ are the **invariant factors** of G .

The order of a finite Abelian group is the product of its invariant factors. G is said to be of type $(n_1, n_2, \dots, n_s)$, and s is the **rank** of G .

E.g. For an Abelian group of order 180, the possible invariant factors are

Invariant Factors	Abelian Groups
$2^2 \cdot 3^2 \cdot 5$	Z_{180}
$2 \cdot 3^2 \cdot 5, 2$	$Z_{90} \times Z_2$
$2^2 \cdot 3 \cdot 5, 3$	$Z_{60} \times Z_3$
$2 \cdot 3 \cdot 5, 2 \cdot 3$	$Z_{30} \times Z_6$

Theorem: Let G be an Abelian group of order $n > 1$ with unique prime factorization

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$$

Then $G \cong A_1 \times A_2 \times \cdots \times A_k$, with $|A_i| = p_i^{\alpha_i}$. For each $A \in \{A_1, \dots, A_k\}$ with $|A| = p^\alpha$,

$$A \cong Z_{p^{\beta_1}} \times Z_{p^{\beta_2}} \times \cdots \times Z_{p^{\beta_t}}$$

where $\beta_1 \geq \beta_2 \geq \dots \geq \beta_t \geq 1$ and $\beta_1 + \beta_2 + \dots + \beta_t = \alpha$. **This decomposition is unique.** p^{β_i} are the **elementary divisors** of G , and the decomposition is the **elementary divisor decomposition** of G .
I.e. G is the direct product of its Sylow p -subgroups.

E.g. For the Abelian group of order $n = 1800 = 2^3 \cdot 3^2 \cdot 5^2$,

Order p^β	Partitions of β	Abelian Groups
2^3	$(3), (2, 1), (1, 1, 1)$	$Z_8, Z_4 \times Z_2, Z_2 \times Z_2 \times Z_2$
3^2	$(2), (1, 1)$	$Z_9, Z_3 \times Z_3$
5^2	$(2), (1, 1)$	$Z_{25}, Z_5 \times Z_5$

Choose 1 Abelian group from each row. E.g. $G \cong Z_8 \times Z_3 \times Z_3 \times Z_{25}$. This **elementary divisor decomposition** can convert to **invariant factor decomposition** and vice versa.

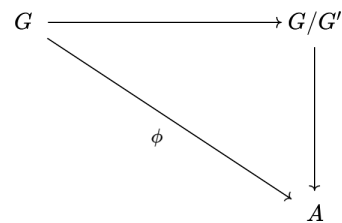
5.3 Recognizing Direct Products

Let $x, y \in G$ and $A, B \subseteq G$ be nonempty subsets. Then $[x, y] = x^{-1}y^{-1}xy$ is called the **commutator** of x and y .

- $[A, B] = \langle [a, b] \mid a \in A, b \in B \rangle$ is the group generated by commutation of elements from A and B .
- $G' = \langle [x, y] \mid x, y \in G \rangle$ is the **commutation subgroup** of G .

Let $x, y \in G$ and $H \leq G$. Then

- $H \trianglelefteq G$ iff. $[H, G'] \leq H$
- $\sigma[x, y] = [\sigma(x), \sigma(y)]$ for any automorphism σ of G .
- If $\phi : G \rightarrow A$ is the homomorphism of G into Abelian group A , then ϕ factors through G' ,



i.e. $G' \leq \ker \phi$. The following diagram commutes: |
 diagram commutes |

| |:-| | This

Recognition Theorem: Let H, K be subgroups of G such that $H, K \trianglelefteq G$ and $H \cap K = 1$. Then

$$HK \cong H \times K$$

The LHS is the **internal direct product** with elements hk . The RHS is the **external direct product** with elements (h, k) .

Part II

Ring Theory

6 Introduction to Rings

6.1 Definitions

A ring is a set R with binary operations $+$, $\times$ satisfying

1. $(R, +)$ is an Abelian group
2. $\times$ is associative, $(a \times b) \times c = a \times (b \times c)$ for all $a, b, c \in R$
3. Distributive law holds:

$$(a + b) \times c = (a \times c) + (b \times c)$$

$$a \times (b + c) = (a \times b) + (a \times c)$$

- If $\times$ is commutative, R is called a **commutative ring**.
- R has identity 1 if there exists $1 \in R$ such that $1 \times a = a \times 1 = a$ for all $a \in R$.
- A subgroup of R closed under multiplication is called a **subring of R** . E.g. $\mathbb{Z}$ is a subring of $\mathbb{Q}$, and $\mathbb{Q}$ is a subring of $\mathbb{R}$.

Let R be a ring. A nonzero $a \in R$ is a **zero divisor** if there exists $b \in R$ such that $ab = 0$ or $ba = 0$.

Let R have identity 1. An element $u \in R$ is a **unit in R** if there exists $v \in R$ such that $uv = vu = 1$. The set of units in R is $R^\times$. Note that zero divisors cannot be a unit.

A ring R with identity $1 \neq 0$ is called a **division ring/skew field** if every nonzero $a \in R$ has a multiplicative inverse. A commutative division ring is called a **field**.

- Or: a field is a commutative ring with identity $1 \neq 0$ in which every nonzero element is a unit.
 - I.e. a field contains no zero divisors. $\mathbb{R}$ is an example of a field, but $\mathbb{Z}$ is not.
-

A commutative ring with identity $1 \neq 0$ and no zero divisors is called an **integral domain**. I.e. rings with same structure as $\mathbb{Z}$.

Theorem: Any finite integral domain is a field.

Proof: Let R be a finite integral domain, with nonzero $a \in R$. Then $x \mapsto ax$ is an injection. Since R is finite, $x \mapsto ax$ is also surjective, and thus there exists $b \in R$ such that $ab = 1$. I.e. a is a unit in R , and since a was arbitrary, R is a field.

6.2 Types of Rings

6.2.1 Polynomial Rings

Let R be a commutative ring with identity. The sum $a_n x^n + \cdots + a_1 x + a_0$ with $a_n \neq 0$ and $n \geq 0$ is the polynomial in x with coefficients $a_i \in R$.

The set of all such polynomials is called the **ring of polynomials** in the variable x with coefficients in R , denoted $R[x]$.

Note that $R \subset R[x]$. R is the subset of constant polynomials.

6.2.2 Matrix Rings

Fix a ring R (doesn't need to be commutative) and let $n \in \mathbb{Z}^+$. Let $M_n(R)$ be the set of all $n \times n$ matrices with entries from R . This is the **matrix ring**.

- If R is nontrivial and $n \geq 2$, then $M_n(R)$ is not commutative.
- Even if R has no zero divisors, $M_n(R)$ with $n \geq 2$ can have zero divisors. Easy to construct.

6.2.3 Group Rings

Fix a commutative ring R with identity $1 \neq 0$ and let $G = \{g_1, g_2, \dots, g_n\}$ be a finite group. **Group rings** RG of G with coefficient in R is the set of sums

$$a_1 g_1 + a_2 g_2 + \cdots + a_n g_n, \quad a_i \in R, \quad 1 \leq i \leq n$$

E.g. for the dihedral group D_8 , the group ring $\mathbb{Z}D_8$ has elements like $r + r^2 - 2s, -3r^2 + rs$, etc.

If $|G| > 1$, then RG always has zero divisors. E.g. let $g \in G$ and $|g| = m$. Then

$$(1 - g)(1 + g + \cdots + g^{m-1}) = 1 - g^m = 0$$

If S is a subring of R , then SG is a subring of RG . Similarly, if H is a subgroup of G , then RH is a subring of RG .

6.3 Ring Homomorphisms

A ring homomorphism is a map from 1 ring to another, respecting the additive and multiplicative structures. Let R, S be rings. $\phi : R \rightarrow S$ is a ring homomorphism if

$$\begin{aligned}\phi(a + b) &= \phi(a) + \phi(b) \\ \phi(ab) &= \phi(a)\phi(b)\end{aligned}$$

The kernel of ϕ is denoted $\ker \phi$, the set of elements of R that map to $0 \in S$.

The homomorphism is called an isomorphism if it's bijective.

- E.g. $\phi : \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$ is a ring homomorphism (respects additive and multiplicative structures).
- E.g. $\phi : \mathbb{Q}[x] \rightarrow \mathbb{Q}$ with $\phi(p(x)) = p(0)$ for $p(x) \in \mathbb{Q}[x]$ is a ring homomorphism.

6.3.1 Ring Ideals

Previously, normal subgroups were the kernels of group homomorphisms. Now, ideals will be kernels of ring homomorphisms.

Let I be an arbitrary subset of the ring R , and $r \in R$. I is the **left ideal** of R if

1. I is a subring of R (closed under $+$, $\times$)
2. I is closed under left multiplication, $rI \subseteq I$ for all $r \in R$.

Right ideals are defined similarly. If I is both a left and right ideal, it's a **two-sided ideal** of R .

Ideals are kernels of ring homomorphisms.

- Fibers of ϕ : $\{r + I \mid r \in R\}$
- The fibers of ϕ have a structure **isomorphic** to the image of ϕ .
- I.e. if X is fiber over $a \in R$ and Y is fiber over $b \in R$, then $X + Y$ is fiber over $a + b$ and XY is fiber over ab .

The ring of cosets, R/I for an ideal I , is called the **quotient ring** of R by $I = \ker \phi$. R/I is a ring under (for $r, s \in R$):

$$\begin{aligned} (r + I) + (s + I) &= (r + s) + I \quad \text{or} \quad \bar{r} + \bar{s} = \overline{r + s} \\ (r + I) \times (s + I) &= (rs) + I \quad \text{or} \quad \bar{r}\bar{s} = \overline{rs} \end{aligned}$$

6.3.2 The Isomorphism Theorems

1st Isomorphism Theorem: If $\phi : R \rightarrow S$ is a homomorphism of rings, then:

1. $\ker \phi$ is an ideal of R .
2. Image of the homomorphism, $\phi(R)$, is a subring of S .
3. $R/\ker \phi \cong \phi(R)$

2nd Isomorphism Theorem: Let A be a subring and B an ideal of R . Then:

1. $A + B$ is a subring of R .
2. $A \cap B$ is an ideal of A .
3. $(A + B)/B \cong A/(A \cap B)$

3rd Isomorphism Theorem: Let I, J be an ideal of R with $I \subseteq J$. Then J/I is an ideal of R/I and

$$(R/I)/(J/I) \cong R/J$$

4th Isomorphism Theorem: Let I be an ideal of R . Define a correspondence $A \leftrightarrow A/I$ between the set of subrings A of R that contain I , and the set of subrings of R/I . *This is a bijection.* A (subring containing I) is an ideal of R iff. A/I is the ideal of R/I .

6.3.3 Solving Diophantine Equations with $\phi : \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$

Note that $\phi : \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ is a ring homomorphism formed by factoring out the ideal $n\mathbb{Z}$. We can use this to solve Diophantine equations.

E.g. $x^2 + y^2 = 3z^2$. This is a relation between elements of the ring $\mathbb{Z}$, thus the relation must also hold in any quotient ring since the structure is preserved. We assume $\gcd(x, y, z) = 1$ since otherwise we can get a more reduced equation.

Consider $\phi : \mathbb{Z} \rightarrow \mathbb{Z}/4\mathbb{Z}$, since the squares $0^2, 1^2, 2^2, 3^2$ are just $0, 1 \pmod{4}$. Thus in $\mathbb{Z}/4\mathbb{Z}$, the equation is

$$\begin{Bmatrix} 0 \\ 1 \end{Bmatrix} + \begin{Bmatrix} 0 \\ 1 \end{Bmatrix} = 3 \begin{Bmatrix} 0 \\ 1 \end{Bmatrix} \equiv \begin{Bmatrix} 0 \\ 3 \end{Bmatrix} \pmod{4}$$

The only solution is $x^2, y^2, z^2 = 0 \pmod{4}$, so by checking choices, they must all be even (0 or 2 mod 4). But then $\gcd(x, y, z) \neq 1$, a contradiction, thus this equation has no solution in $\mathbb{Z}$.

6.4 Properties of Ideals

6.4.1 Finitely Generated Ideals

Let A be a subset of a ring R . (A) denotes the smallest ideal of R containing A and is called the **ideal generated by A** .

- Principal ideal: ideal generated by 1 element
- Finitely generated ideal: ideal generated by finite set

$$(A) = \bigcap_{I \text{ an ideal}, A \subseteq I} I$$

E.g. $\{0\}$ and R are principal ideals: $0 = (0)$ and $R = (1)$.

E.g. $n\mathbb{Z}$ is a principal idea: $n\mathbb{Z} = (n)$.

6.4.2 Maximal Ideals

Theorem: Let I be an ideal of R .

- 1) $I = R$ iff. I contains a unit.
- 2) If R is commutative, R is a field iff its only ideals are 0 and R .

Proof: (1) If $I = R$, I contains unit 1. If u is a unit in I with inverse v , for any $r \in R$,

$$r = r \cdot 1 = r(vu) = (rv)u \in I$$

- (2) If R is a field, every nonzero element must be a unit, i.e. every nonzero ideal contains a unit. By part (1), R is the only nonzero ideal. Other direction: if $0, R$ are only ideals of R , let $u \in R$, $u \neq 0$. R is commutative, so $R = (u)$ and $1 \in (u)$, i.e. $1 = vu$, i.e. u is a unit. u is arbitrary, so R is a field.

An ideal M in arbitrary ring R is called a **maximal ideal** if $M \neq R$ and the only ideals containing M are M and R .

- In a ring with identity, every proper ideal is contained in a maximal ideal.
- If R is commutative, the ideal M is maximal iff. R/M is a field. (proof: 4th isomorphism theorem + previous theorem)

Above gives a procedure for constructing fields: 1) Get a ring, 2) Divide by maximal ideal. E.g. $n\mathbb{Z}$ is a maximal ideal of $\mathbb{Z}$ iff. $\mathbb{Z}/n\mathbb{Z}$ is a field, i.e. n is prime.

6.4.3 Prime Ideals

Let R be a commutative ring. Ideal P is a **prime ideal** if $P \neq R$ and whenever $ab \in P$ for $a, b \in R$, at least one of a, b is in P .

E.g. ideal $n\mathbb{Z}$ with $n > 1$ is a prime ideal of $\mathbb{Z}$ iff. for every time $ab \in n\mathbb{Z}$ for $a, b \in \mathbb{Z}$, at least 1 of a, b is in $n\mathbb{Z}$. I.e. whenever n divides ab , n divides either a or b or both. I.e. n is prime — and $p\mathbb{Z}$ are the prime ideals of $\mathbb{Z}$.

0 can also be a prime ideal.

Theorem: If R is commutative, the ideal P is a prime ideal in R iff. R/P is an integral domain.

Proof: P is prime iff. whenever $ab \in P$, either $a \in P$ or $b \in P$. Let $\bar{r} = r + P$. Then $r \in P$ iff. $\bar{r} = 0$ in R/P . Thus P is a prime ideal iff. whenever $\bar{a}\bar{b} = \bar{0}$, $\bar{a} = 0$ or $\bar{b} = 0$. I.e. R/P is an integral domain, no zero divisors.

Theorem: If R is commutative, every maximal ideal of R is a prime ideal.

Proof: If M is a maximal ideal, R/M is a field by previous theorems. Every field is an integral domain, thus M is a prime ideal.

- E.g. $p\mathbb{Z}$ is both a prime and maximal ideal in $\mathbb{Z}$.
- Note that above doesn't work in reverse. E.g. (x) is a prime ideal in $\mathbb{Z}[x]$ since $\mathbb{Z}[x]/(x) \cong \mathbb{Z}$, an integral domain. But this is not a field, so (x) is not a maximal ideal in $\mathbb{Z}[x]$.
- The problem is that not all “integral domains” are “fields” (but field = finite integral domain works).

6.5 Rings of Fractions

Let R be a commutative ring. Let D be a nonempty subset of R that doesn't contain zero or any zero divisors, and is closed under multiplication. Then there exists a commutative ring Q with $1 \in Q$ such that R is a subring of Q and every element of D is a unit in Q .

- Every element of Q is of the form rd^{-1} for $r \in R$ and $d \in D$.
- If $D = R - \{0\}$, then Q is a field.
- Q is unique up to an isomorphism.

Q is called a “**ring of fractions**” of D w.r.t R . Denoted $Q = D^{-1}R$. Essentially, the smallest field that can be built by taking ratios of integral domains.

If R is an integral domain and $D = R - \{0\}$, then Q is a **field of fractions** or **quotient field** of R .

- E.g. if R is a field, its field of fractions is just R .
- The field of fractions of $\mathbb{Z}$ is $\mathbb{Q}$.
- The field of fractions of $2\mathbb{Z} \subset \mathbb{Z}$ is $\mathbb{Q}$.

6.6 Chinese Remainder Theorem

A ring direct product is $R_1 \times R_2$ with ordered pairs (r_1, r_2) such that

$$\begin{aligned}(r_1, r_2) + (s_1, s_2) &= (r_1 + s_1, r_2 + s_2) \\ (r_1, r_2)(s_1, s_2) &= (r_1 s_1, r_2 s_2)\end{aligned}$$

Ideals A, B of R are called **comaximal** if $A + B = R$. Then $A \cap B = AB$.

Chinese Remainder Theorem: Let $A_1, A_2, \dots, A_k$ be ideals in R . The map

$$R \rightarrow R/A_1 \times R/A_2 \times \dots \times R/A_k$$

defined by $r \mapsto (r + A_1, r + A_2, \dots, r + A_k)$ is a ring homomorphism with kernel $A_1 \cap A_2 \cap \dots \cap A_k$. If A_i, A_j are comaximal for every $i \neq j$, then $A_1 \cap A_2 \cap \dots \cap A_k = A_1 A_2 \dots A_k$. Thus

$$R/(A_1 A_2 \dots A_k) \cong R/A_1 \times R/A_2 \times \dots \times R/A_k$$

The special case is $\mathbb{Z}/mn\mathbb{Z} \cong (\mathbb{Z}/m\mathbb{Z}) \times (\mathbb{Z}/n\mathbb{Z})$ for coprime m, n .

Let $n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k}$ for prime p_i . Then

$$(\mathbb{Z}/n\mathbb{Z}) \cong (\mathbb{Z}/p_1^{\alpha_1}\mathbb{Z}) \times (\mathbb{Z}/p_2^{\alpha_2}\mathbb{Z}) \times \dots \times (\mathbb{Z}/p_k^{\alpha_k}\mathbb{Z})$$

This is an isomorphism of rings, so a group of units should also obey the isomorphism. Thus

$$\begin{aligned}(\mathbb{Z}/n\mathbb{Z})^\times &\cong (\mathbb{Z}/p_1^{\alpha_1}\mathbb{Z})^\times \times (\mathbb{Z}/p_2^{\alpha_2}\mathbb{Z})^\times \times \dots \times (\mathbb{Z}/p_k^{\alpha_k}\mathbb{Z})^\times \\ \phi(n) &= \phi(p_1^{\alpha_1})\phi(p_2^{\alpha_2}) \dots \phi(p_k^{\alpha_k})\end{aligned}$$

7 Domains, Domains, Domains

7.1 Euclidean Domains

7.1.1 Definitions

We define a norm as a measure of “size” in rings. A **norm** is any function $N : R \rightarrow \mathbb{Z}^+ \cup \{0\}$ with $N(0) = 0$. If $N(a) > 0$ for all $a \neq 0$, then N is a **positive norm**. This definition is very loose.

An integral domain R is a **Euclidean domain** if there exists a norm N on R such that for any $a, b \in R$ with $b \neq 0$, there exists $q, r \in R$ such that

$$\begin{aligned} a &= qb + r, \quad r = 0 \text{ or } N(r) < N(b) \\ q &= \text{quotient} \\ r &= \text{remainder} \end{aligned}$$

I.e. R is a Euclidean domain if there exists a division algorithm.

- $\mathbb{Z}$ is a Euclidean domain with norm $N(a) = |a|$.
- If F is a field, $F[x]$ is a Euclidean domain with norm $N(p(x)) = \text{degree of } p(x)$.

Theorem: Every ideal of a Euclidean domain is principal. I.e. if I is any nonzero ideal in a Euclidean domain R , then $I = (d)$ where d is a nonzero, minimum-norm element of I .

Proof: Let d be a nonzero element of I with minimum norm, which exists by well-ordering of $\mathbb{Z}$. Then $(d) \subseteq I$. Conversely, let $a \in I$ and use the division algorithm to write $a = qd + r$ with $N(r) < N(d)$. Then $r = a - qd$. Both a, qd are in I by definition, so $r \in I$. Since $N(r) < N(d)$ and $N(d)$ is minimal, $r = 0$. Thus $a = qd \in (d)$, i.e. $I \subseteq (d)$.

- E.g. Every ideal $n\mathbb{Z}$ of the Euclidean domain $\mathbb{Z}$ are principal. $n\mathbb{Z} = (n)$.
- The ring of polynomials $\mathbb{Q}[x]$ is a Euclidean domain since $\mathbb{Q}$ is a field.
- But the ring of polynomials $\mathbb{Z}[x]$ is not a Euclidean domain since $\mathbb{Z}$ is not a field. Or: $(2, x)$ is an ideal of $\mathbb{Z}[x]$ that is not principal, hence by above, $\mathbb{Z}[x]$ cannot be a Euclidean domain.

7.1.2 Greatest Common Divisors

A Euclidean domain provides the notion of a GCD. If I is an ideal generated by a, b and $I = (d)$ where the norm of d is maximal, then d is the GCD of a, b .

The GCD can be computed algorithmically. Let R be the Euclidean domain with $a, b \in R$ and $a, b \neq 0$. Following the **Euclidean algorithm**,

$$\begin{aligned} a &= q_0b + r_0 \\ b &= q_1r_0 + r_1 \\ r_0 &= q_2r_1 + r_2 \\ &\vdots \\ r_{n-2} &= q_nr_{n-1} + r_n \\ r_{n-1} &= q_{n+1}r_n \end{aligned}$$

Then $\gcd(a, b) = r_n$. This is a very fast algorithm.

7.2 Principal Ideal Domains

A **principal ideal domain** is an integral domain in which every ideal is principal.

Theorem: Every nonzero prime ideal in a PID is a maximal ideal.

Proof: Let (p) be a nonzero prime ideal in R , and let $I = (m)$ be an ideal containing (p) . We seek to show that either $I = (p)$ or $I = R$. Since $p \in (m)$, $p = rm$ for some $r \in R$. (p) is a prime ideal, so either $r \in (p)$ or $m \in (p)$.

- If $r \in (p)$, then $r = ps$, $p = rm = psm$ and $sm = 1$. m is a unit, thus $I = R$.
- If $m \in (p)$, then $(m) = (p) = I$.

Theorem: If R is a commutative ring such that $R[x]$ is a PID, then R is a field.

Proof: If $R[x]$ is a PID, then R is an integral domain. Since $R[x]/(x) \cong R$, (x) is a nonzero prime ideal in R . By the previous theorem, (x) is a maximal ideal, thus R is a field.

7.2.1 Dedekind-Hasse Norm

N is a **Dedekind-Hasse norm** if

- 1) N is a positive norm
- 2) For all $a, b \in R$, either $a \in (b)$ or $\exists s, t \in R$ such that $0 < N(sa - tb) < N(b)$.

This is a weak version of the Euclidean condition for Euclidean domains.

An integral domain is a PID iff. it has a Dedekind-Hasse norm.

- Thus not every PID is an ED (Dedekind-Hasse norm $\neq$ Euclidean norm)
- But every ED is a PID (Euclidean norms satisfy Dedekind-Hasse norm, e.g. $s = 1$)

7.3 Unique Factorization Domains

8 Polynomial Rings